

Technology_Selection_ADR_Standard

Technical Standard

保险公司 技术开发部
内部文档 · 仅供授权人员查阅

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Technology Selection & ADR Standard

1. Overview

This document defines the technology selection process and Architecture Decision Record (ADR) standards, ensuring transparent, traceable, and reviewable technical decisions. References include ThoughtWorks Tech Radar and Netflix engineering practices.

2. Technology Selection

2.1 Triggers

- New project/system requires technology stack
- Existing stack reaches EOL or unmaintained
- Business requirements change rendering current solution inadequate
- Performance/cost optimization needed
- Security vulnerabilities requiring component replacement

2.2 Evaluation Dimensions

Dimension	Weight	Assessment
Functional Fit	25%	Core features, extensibility
Community Activity	15%	Stars, Commits, Contributors, Release frequency
Maturity	15%	Production cases, stability, commercial support
Team Familiarity	10%	Existing skills, learning curve
Ecosystem Fit	10%	Integration difficulty with existing stack
Performance & Scalability	10%	Benchmarks, horizontal scaling
License & Cost	10%	Open source license, commercial cost, ops cost
Security	5%	CVE history, security response speed

2.3 Selection Process

Requirements → Candidate Research → PoC → Score & Decision → Team Review → Adoption

- Minimum 3 candidates per selection
- PoC must cover core functionality and performance scenarios
- Decision recorded as ADR

3. Architecture Decision Records

3.1 ADR Template

```
# ADR-NNN: Title
## Status
[Proposed | Accepted | Deprecated | Superseded]
## Context
Decision context and problem to solve
## Options
### Option A
- Pros:
```

```

- Cons:
### Option B
- Pros:
- Cons:
## Decision
Option A
Rationale:
1. ...
2. ...
## Consequences
Positive and negative implications
## Related
- Related ADRs: ADR-001, ADR-012
- Related docs: Microservice Architecture Standard

```

3.2 ADR Management

- Stored in source repo /docs/adr/
- Sequentially numbered, never delete or modify accepted ADRs
- Each ADR includes clear context, options, and rationale
- New ADR supersedes old when better approach emerges
- Team review required before marking "Accepted"

4. Technology Stack Inventory

Category	Adopted	Restricted	Retired
Backend	Spring Boot 3.x, Spring Cloud	Quarkus, Micronaut	Legacy Spring MVC
Frontend	React 18, Vue 3	Angular	JSP
Database	MySQL 8.0, Redis, ES	PostgreSQL	Oracle
Messaging	RocketMQ	Kafka, RabbitMQ	ActiveMQ
Container	Docker, Kubernetes	-	Docker Swarm
CI/CD	GitLab CI, ArgoCD	Jenkins	-
Monitoring	Prometheus, Grafana	Datadog	Zabbix
Languages	Java 17, Python 3.11, TypeScript	Go	-